

EMHD Nanofluid Flow Along A Porous Riga Plate With Thermal Radiation

Mehetaj Parvine^{†,*}, Md Tusher Mollah^{‡,††}, Saykat Poddar^{‡‡}, Md. Mahmud Alam^{††‡}, and Giulio Lorenzini^{††‡‡}

[†] Department of Mathematics, Patuakhali Science and Technology University, Dumki 8602, Patuakhali, Bangladesh.

[‡] Department of Mechanical Engineering, Technical University of Denmark, Kgs. Lyngby 2800, Denmark.

^{††} Department of Mathematics, European University of Bangladesh, Dhaka 1216, Bangladesh.

^{‡‡} Department of Mathematics, Faculty of Science, Bangabandhu Sheikh Mujibur Rahman Science and Technology University, Gopalganj 8100, Bangladesh.

^{††‡} Mathematics Discipline, Science, Engineering and Technology School, Khulna University, Khulna 9208, Bangladesh.

^{††‡‡} Department of Engineering and Architecture, University of Parma, Parco Area delle Scienze 181/A, Parma 43124, Italy.

*Corresponding Author's Email: mehetajparvine@pstu.ac.bd; mehetajparvine@gmail.com

ABSTRACT

This article pays attention to the numerical analysis of unsteady electro-magnetohydrodynamic (EMHD) Nanofluidic flow past a Riga plate of porosity criteria with thermal radiation. The flow is characterized under the assumption of radiation phenomena. Using the usual transformations, the non-dimensional mathematical model in the form of nonlinear coupled partial differential equations is renewed from the dimensional form. The interactions of thermophoretic and Brownian features for nanofluid are analyzed. The associated governing equations for the participated boundary settings are discretized and reckoned with by the explicit finite difference method (EFDM). The flow momentum, temperature, and concentration profiles along with shear stress, Nusselt number, and Sherwood number have been discussed graphically for sundry logical values of some important parameters. Also, to authenticate the novelty of this article, the possible sensitivities, and criteria of solutions' convergence along with stability have been demonstrated graphically and statistically.

KEYWORDS

Riga plate, Electro-magnetohydrodynamic field, Nanofluid, Porous medium, Numerical Analysis

INTRODUCTION

In recent decades, the study of electromagnetic Nanofluid flow has been achieving considerable attraction of researchers for being used extensively in industrial manufacturing and engineering processes like the imposition of Riga plate to the ship and submarine industry. The property of better thermal conductivity of nanofluids leads the investigators to intensify the rigorous analysis of nanofluids in microscale heat transfer, nuclear reactors, and biomedicine, etc. With regard to the multiple applications of nanofluid in scientific and industrial sectors, the research on nanofluid is gaining worldwide attention from various researchers in recent years. Choi [1] was the pioneer proposing the concept of nanofluid. Many authors have studied Nanofluid flows over or past Riga plates to contribute to the development of flow control devices in the submarine industry, reducing submarines' friction force as well as pressure drag by ensuring the prevention of boundary-layer separation. Riga plate, the monitoring device of flow, was incorporated by Gailitis and Lielausis [2]. Generation of the "Grinberg term" for the Newtonian fluid occurs that consequences from the imposition of the Riga plate, which later introduces the modified Hartmann number [3] for replacement of Hartmann number [4]. The resulting parameter happens due to the parallel Lorentz force throughout the boundary layer.

Positive assumption of the parameter reveals the forward flow criteria, while backflow is observed for the negative quantities [5]. After the time of detecting Lorentz force under the presence of the actuator, the investigators have been captivated in researching with Riga plate. Abbas et al. [6] examined the suction case of entropy production in nanofluid on a horizontally ascended Riga plate. Wang and Hayat [7] studied Maxwell fluid for the fluctuating nature under variable suction control. Their investigation decided that the more the oscillating amplitude, the more the flow, albeit the flow outside the boundary layer possesses forward-flowing nature. Ahmad et al. [8] took an effort to analyze the asymptotic nature of nanofluid flow compared to the numerical reckoning. They performed linear regression to correlating the emerging physical quantities. Hayat et al. [9] conducted a comparative numerical investigation of magnetized squeezing flow with the Riga plate, mainly focusing on the blanket flow. The fluid temperature is reported to be developing for the existence of the radiative phenomena. Iqbal et al. [10] considered an accelerating Riga plate to establish the bio convection for the motility of gyrotactic microorganisms. They concluded rising statistics in velocity for the increasing functioning of microorganism concentration. Mollah et al. [11] executed a worthwhile investigation on the complex rheological features of Bingham materials under the action of Hall current and ion-slip prescription.

The fluid temperature was noted faster to reach stability in time over unsteadiness, compared to the velocities. An extensive numerical performance on the thixotropic nano liquid for biotechnological uses has been researched by Khan et al. [12]. The fluid motion was reported to hike up for the implication of rising values of thixotropic parameters. Khatun et al. [13] investigated the radiating fluid with the Riga plate and reported suppression on the velocities for suction. Iterative shooting procedure was employed by Ramzan et al. [14] to examine the features of thermophoretic and Brownian actions on radiatively responsive nanofluid of Williamson type. Rasool et al. [15] deliberated the Eyring-Powell method of non-Newtonian rheology for nanofluid under convective boundaries. Deferment of nanoparticles leads the flow to be intensified along with the mass-heat flux. Shamshuddin et al. [16] went for the VPM analytical approach to quest the squeezed flow concerning Ohmic generalization of the widely used Fourier law was prescribed with relaxation time. They targeted to develop a model that would aid the sensor designing along with the smarter biomechanical lubrication. Nassem et al. [17] applied the Cattaneo-Christov method to deal with the nanofluid flow by incorporating the homotopy solution procedure. The findings concluded an interesting observation that the relaxation time reveals suppressing impact on the third-grade fluid temperature and concentration.

Ramesh and Giresha [18] constructed a radiative nanofluid flow problem that could be converted to the Blasius and Sakiadis structure under reasonable restrictions. The classical problems resulted in a bosting in velocity due to the Riga plate familiarization. Shafiq et al. [19] applied the homotopy analytical mechanism on the Marangoni flow embedding in the Riga plate to compare the nanoparticles' heat fluxes for the magnetized force. Anjum et al. [20] described flow on the stagnation point for a non-linearly thicked Riga surface along with thermal slips. They have noted that the flow velocity suppresses while the wall thickness develops. Abbas et al. [21] analyzed the Electromagnetohydrodynamic flow of nanofluidic metarials through a porous Riga surface by considering the gyrotactic microorganisms. In 2020, a practical assessment on Bioconvection in Maxwell's nanofluidic flow was outlined by a Riga surface with the response of radiation along with the workings of activation energy by Ramesh et al. [22]. A comprehensive experiment on Transient electromagnetohydrodynamic and destructively specified reactive flow between a couple of convectively heated actuators for a squeezing nature has been researched by Thumma et al. [23]. Loganathan et al. [24] developed a boundary layer control model using radiative Casson fluid flow over a permeable Riga plate for identifying Unimolecular Chemical Reaction of distribution.

Mollah [25] formed a worthwhile investigation on the laminar flow of EMHD Bingham elements within a couple of parallel Riga plates with horizontal orientation. Comparative numerical and scale analysis of the nanofluidic flow of Eyring-Powell type through a radiative Riga wall by taking the impacts of lubrication into account has been executed by Chu et al. [26]. Ansari et al. [27] applied the spectral quasilinearization accession to deal with the Jeffrey nanofluid flow close to the Riga plate. There is another recent statistical exploration of MHD heat and mass transfer dissipative with radiating fluid flow with suction purposes resulted by Poddar et al. [28]. Drawing motivation from the prior research works, the current article pays to heed the numerical analysis of flow nature for EMHD Nanofluid over a porous and vertically subjected Riga plate with the workings of the radiative phenomena. The constituted equations of the flow problem have been functionalized by the precise

explicit finite difference method (EFDM). The consequences of these computations are manifested in graphs for nanofluid concentration, temperature, and velocity within the framework of the emerging parameters.

MATHEMATICAL MODEL OF THE FLOW

This analysis deals with the transient nanofluid flow under the influence of a vertical Riga surface with the impact of thermal radiation. The alternating sequence of permanent electrodes and magnets are amalgamated and mounted vertically on a plane to constitute a Riga plate (Fig. 1). The plan for the structured flow is modeled as demonstrated in Fig.2. The flow direction is upward subjected along the Riga surface i.e., in the x-direction, while the y-axis is deliberated to be in the normal direction to the plate. For the Riga plate, a wall parallel Lorentz force can be obtained from the operation of current density and magnetic induction that can be expressed as $\mathbf{F} = \mathbf{J} \wedge \mathbf{B}$, as shown in (Fig.1). Grinberg first showed this $\mathbf{F}$. The average force density along the spanwise coordinate z is, $F = F_{ex}$ that comes out as an exponentially decreasing function of y and can be expressed as $F = \mathbf{J} \wedge \mathbf{B} = \frac{\pi}{8} J_0 M_0 \exp(-\frac{\pi}{a} Y)$ where, $J(A/m^2)$ specifies the applied current density of electrodes, M_0 (Tesla) specifies the magnetization of the permanent magnets, a specifies the width of the electrodes and magnets. The temperature T , and nanoparticle concentration C possess constant estimations at the wall (plate) as T_w and C_w respectively. The temperature and nanoparticle concentrations in the outer zone of the boundary layer are marked by T_∞ and C_∞ .

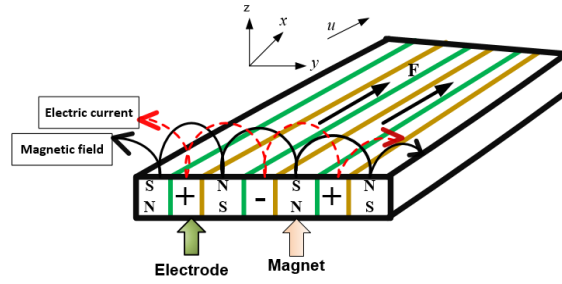


Figure 1. Physical configuration of the Riga plate with the coordinate specification

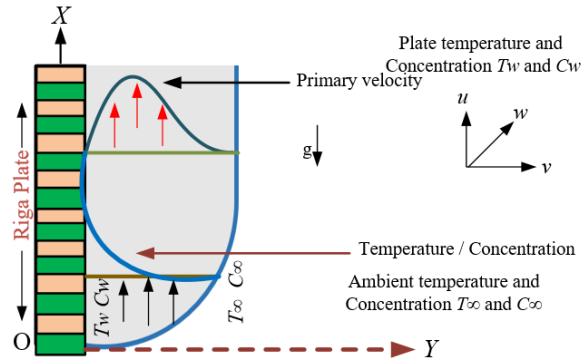


Figure 2. Structure and coordinates of the flow model

The radiative expression of Rosseland can be expressed as follows (Chauhan and Khemchandani [29]):

$$Q_r = -\frac{4\sigma^*}{3k^*} \left(\frac{\partial T^4}{\partial y} \right) \quad (1)$$

Where the mean absorption coefficient is (k^*) , radiative heat flux is (Q_r) , and Stefan-Boltzmann constant is (σ^*) . The well-established expansion (Hassan et al. [30]) of T^4 in Taylor series about T_∞ transforms into $T^4 = 4T_\infty^3 - 3T_\infty^4$, which ultimately converts the above form as:

$$Q_r = -\frac{16\sigma^*}{3k^*} T_\infty^3 \left(\frac{\partial T}{\partial y} \right) \quad (2)$$

The non-dimensionalization of the governing equations has been performed by employing the following quantities:

$$\left. \begin{aligned} X = \frac{x}{l}, Y = \frac{y}{L}, U = \frac{u}{U_w}, V = \frac{v}{V_0}, \theta = \frac{T - T_\infty}{T_w - T_\infty}, \phi = \frac{C - C_\infty}{C_w - C_\infty}, \\ L = \frac{a}{\pi}, l = \frac{U_w L^2}{\nu}, V_0 = \frac{\nu \pi}{a}, \tau = \frac{t U_w}{l} \end{aligned} \right\} \quad (3)$$

The dimensionless governing nonlinear coupled PDEs for current flow mechanism are as follows ([13], [25]):

Continuity equation:

$$\frac{\partial U}{\partial X} + \frac{\partial V}{\partial Y} = 0 \quad (4)$$

Momentum equation:

$$\frac{\partial U}{\partial \tau} + U \frac{\partial U}{\partial X} + V \frac{\partial U}{\partial Y} = \frac{\partial^2 U}{\partial Y^2} + R_c \theta - N_r \phi + H_a e^{-Y} - P_m U \quad (5)$$

Energy equation:

$$\frac{\partial \theta}{\partial \tau} + U \frac{\partial \theta}{\partial X} + V \frac{\partial \theta}{\partial Y} = \left(\frac{1}{Pr} + \frac{4}{3} R_d \right) \frac{\partial^2 \theta}{\partial Y^2} + N_b \frac{\partial \theta}{\partial Y} \frac{\partial \phi}{\partial Y} + N_t \left(\frac{\partial \theta}{\partial Y} \right)^2 \quad (6)$$

Concentration equation:

$$\frac{\partial \phi}{\partial \tau} + U \frac{\partial \phi}{\partial X} + V \frac{\partial \phi}{\partial Y} = \frac{1}{Le} \frac{\partial^2 \phi}{\partial Y^2} + \frac{1}{Le} \frac{N_t}{N_b} \frac{\partial^2 \theta}{\partial Y^2} \quad (7)$$

The dimensionless conditions for initial and boundary settings are as follows:

$$\left. \begin{aligned} U = 0, V = 0, \theta = 1, \phi = 1 \quad \text{at } Y = 0 \\ U \rightarrow 0, V = 0, \theta = 0, \phi = 0 \quad \text{at } Y \rightarrow \infty \end{aligned} \right\} \quad (8)$$

Substitution of non-dimensional quantities in the governing equations lead this study to have sundry dimensionless parameters as follows:

$$\text{Richardson number } R_c = \frac{a^2 \Delta T}{\pi^2 \nu U_w} (1 - C_\infty) g \beta^*$$

$$\text{nanoparticle concentration flux } N_r = \frac{a^2 \Delta C}{\pi^2 \nu U_w} g \frac{\rho_p - \rho_f}{\rho_f},$$

Hartmann number $H_a = \frac{1}{8\pi} \frac{a^2 J_0 M_0}{v U_w \rho_f}$, permeability of porous medium $P_m = \frac{v l}{k' U_w}$, Prandtl number $P_r = \frac{v}{\alpha}$, radiation parameter $R_d = \frac{4\sigma^* T_\infty^3}{k k^*}$, Brownian motion parameter $N_b = \frac{\tau D_B \Delta C}{v}$, Lewis Number $L_e = \frac{v}{D_B}$, and thermophoresis parameter $N_t = \frac{\tau D_t \Delta T}{v T_\infty}$.

Numerical technique

The governing dimensionless coupled PDEs (1)-(7) have been manipulated in the form of difference equations with the use of EFDM regarding the boundary conditions to reckon the numerical values of the solutions. The two-dimensional region inside the boundary layer is set by several grids of lines that are oriented in the parallel direction of X and Y axes where the plate direction is specified by X -axis, and the normal direction of the plate is specified by Y -axis. This configuration is structured in Fig. 3 with the general notation of mesh points. The height of X direction i.e., the plate is conveniently set as, $X_{\max} (= 60)$, which indicates that the value of X varies from 0 to 60, and the perpendicular region's width is thought of, $Y_{\max} (= 25)$, i.e., Y varies from 0 to 25 while $Y \rightarrow \infty$ indicates the outer boundary layer zone. Discretization of the flow zone is done in such a manner that the mesh sizes become constant values and the assumption is noted as ΔX and ΔY in X and Y directions respectively, whereas the calculated values are, $\Delta X = 1.00 (0 \leq x \leq 60)$, $\Delta Y = 0.4 (0 \leq y \leq 25)$ with the smaller time-step, $\Delta \tau = 0.025$

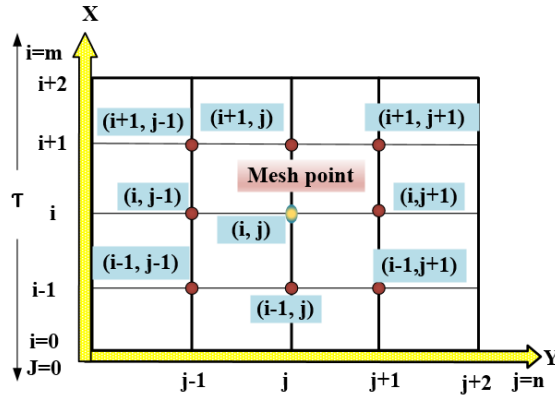


Figure 3. Finite difference grids

Finally, the conditions for the solutions of the current EFDM to be stable are mentioned as follows:

$$\frac{U \Delta \tau}{\Delta X} + |V| \frac{\Delta \tau}{\Delta Y} + 2 \left(\frac{1}{P_r} + \frac{4}{3} R_d \right) \frac{\Delta \tau}{(\Delta Y)^2} + 2 N_b \phi \frac{\Delta \tau}{(\Delta Y)^2} + 2 N_t \theta \frac{\Delta \tau}{(\Delta Y)^2} \leq 1 \quad \frac{U \Delta \tau}{\Delta X} + |V| \frac{\Delta \tau}{\Delta Y} + \frac{2}{L_e} \frac{\Delta \tau}{(\Delta Y)^2} \leq 1$$

Now, the suitable values of meshes i.e. $\Delta X = 1.00 (0 \leq x \leq 60)$, and $\Delta Y = 0.4 (0 \leq y \leq 25)$ generate converging criteria for the solutions as,

$$R_d \leq 1.50, L_e \geq 0.512, P_r \geq 1.00, N_b \leq 7.8125, N_t \leq 7.92$$

RESULTS AND DISCUSSIONS

The retrieved governing equations are thoroughly analyzed with the actions of pertinent parameters on them to unfold the inner mechanical construction of the established physical flow model. As a computational tool to reckon the numerical values at steady-state flow behavior, MATLAB R2015a has been rigorously used. The features of non-dimensional primary velocity (U), temperature (θ), and concentration (ϕ) distributions are

recorded and portrayed graphically. All the possible sensitivities for the EFDM schemes are deliberately analyzed to add an additional factor to the accuracy of the solutions.

Mesh sensitivity test: The best fit for mesh space is a prime concern before reckoning the numerical values. Different pairs of (m, n) values are tested such as $(m, n) = (40, 40; 50, 50; 60, 60)$ which is demonstrated in Fig. 4 . Expected smoothness from the resulting curves motivated to select one of them as the agent of further computations. Consequently, the mesh size $(m, n) = (60, 60)$ is taken as the number of meshes.

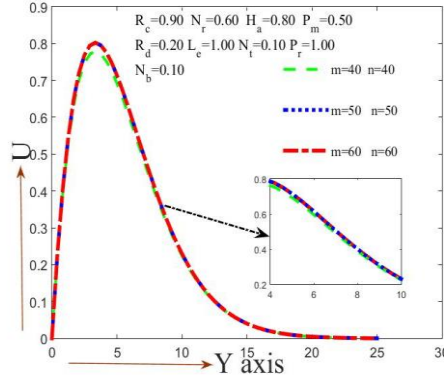


Figure 4. Illustration of mesh sensitivity for velocity profile

Time sensitivity test: At a certain point of time, the unsteadiness of the solutions quit, and after that time-domain literally has no impact that could be able to alter the values of the solutions anymore. This situation is referred to as the steady-state situation. The current study aims to analyze the solutions for the steady-state time value. The steady-state situation for the solutions is searched with several testing quantities of dimensionless time for the velocity, temperature, and concentration profiles. The observation on the testing values reveals that after 20, the variations are not that significant and discloses negligible variations after 40. Therefore, the solutions at $\tau = 40$ are fixed as the steady-state solutions. Fig. 5 shows that the primary velocity attains a gradual steady-state.

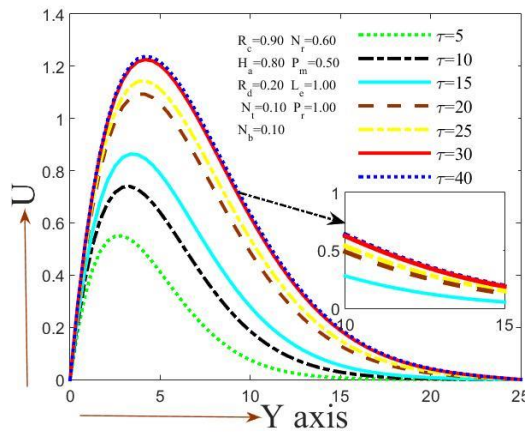


Figure 5. Illustration of time sensitivity for velocity profile

Effect of parameters: The impacts of two different parameters, namely radiation parameter R_d , and Lewis number L_e with the general existence of $R_c = 0.90$, $N_r = 0.60$, $N_t = 0.10$, $H_a = 0.80$, $P_m = 0.50$, $N_b = 0.10$, $P_r = 1.00$ are presented graphically through Figs. 6-14. It is identified from Figs. 6 and 8 that the velocity, and the average shear stress increase for the enhancing values of R_d , and L_e whereas in Fig. 7, local shear stress

decreases for the enhancing values of R_d and rises for L_e . Figs. 9-11 show the effects of temperature, and the local and average Nusselt number for radiation parameter R_d , and Lewis number L_e . In Fig. 9 and 11, temperature profile and average Nusselt number decrease for L_e and increase for R_d , on the other hand in Fig. 10, local Nusselt number increases for both parameters R_d , and L_e . In Figs. 12 -14 the illustration of R_d and L_e on concentration, and the average and local Sherwood number has been given. It is observed from the figs. 12 and 14 that both the concentration distribution and average Sherwood number fall with the expanded values of R_d and L_e while in Fig. 13, it can be observed that the local Sherwood number tends down for rising values of R_d and increases with the increase of L_e .

The simultaneous impact of porous medium parameter P_m , and Brownian motion parameter N_b on velocity, local shear stress, and average shear stress has been depicted in Figs. 15-17. All the mentioned profiles for Figs. 15-17 develops in quantity for the higher values of P_m while N_b makes these profiles shorter.

Figs. 18-20 demonstrate the behavior of temperature, and the average and local Nusselt number for Brownian motion parameter N_b , and thermophoresis parameter N_t . In Fig. 18, a substantial increase with N_b and a gradual increase with N_t is visible for temperature profile. From Figs. 19 and 20, it can be reckoned that both the local and average Nusselt number exhibit tiny decrement for N_b and relatively significant decrement for N_t .

Figs. 21-23 illustrate the behavior of nanofluid concentration, local Sherwood number, and average Sherwood number for Brownian motion parameter N_b , and thermophoresis parameter N_t . From Fig. 21, the fluid concentration falls near the plate with the growth of N_b and develops in the far boundary layer area while the enhancement of N_t is the reason behind the concentration suppression within the whole area. Fig. 22 points out that the local Sherwood number falls with the development of both N_b , and N_t . As mentioned in Fig. 23, the average Sherwood number initially rises (approximately for $0 \leq \tau \leq 4$) and thereafter falls with the rising estimation of N_b while N_t stands to suppress the average Sherwood number gradually.

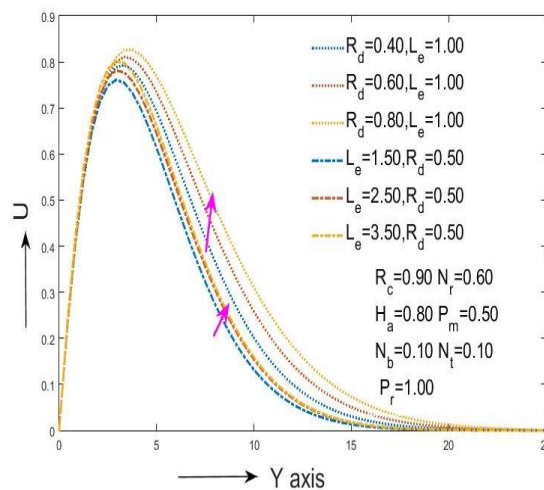


Figure 6. Effect of R_d , and L_e on primary velocity

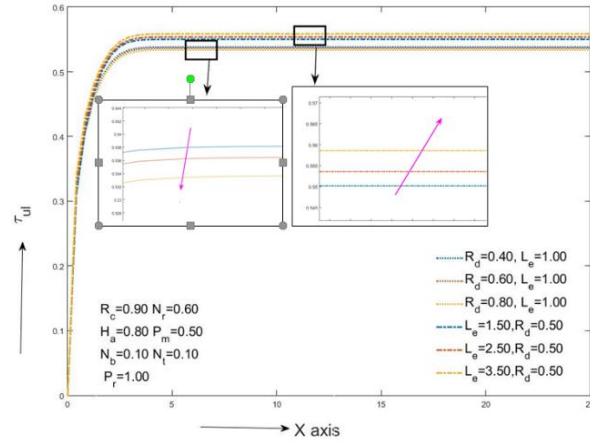


Figure 7. Effect of R_d , and L_e on local shear stress

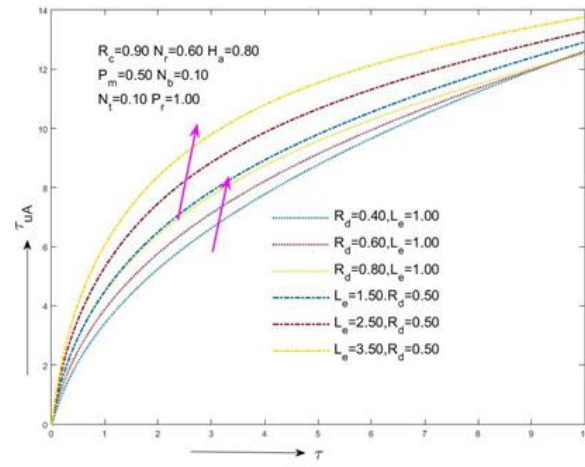


Figure 8. Effect of R_d , and L_e on average shear stress

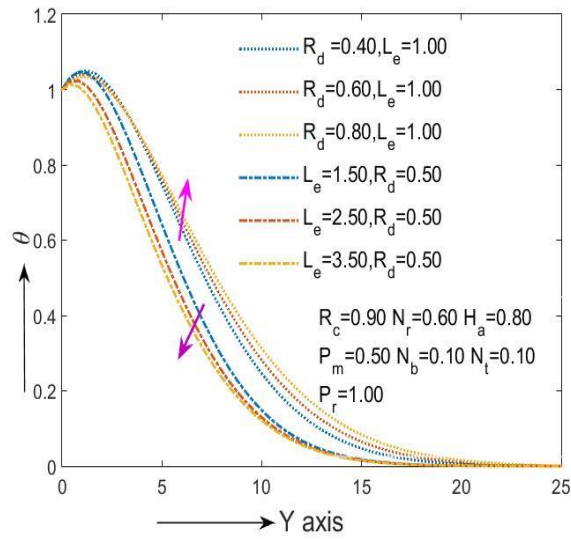


Figure 9. Effect of R_d , and L_e on temperature distribution

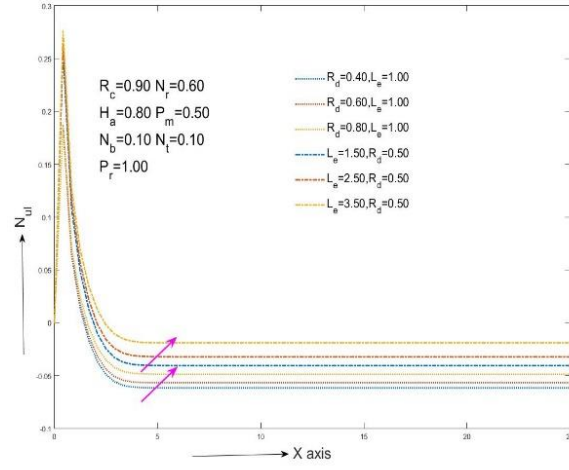


Figure 10. Effect of R_d , and L_e on local Nusselt number

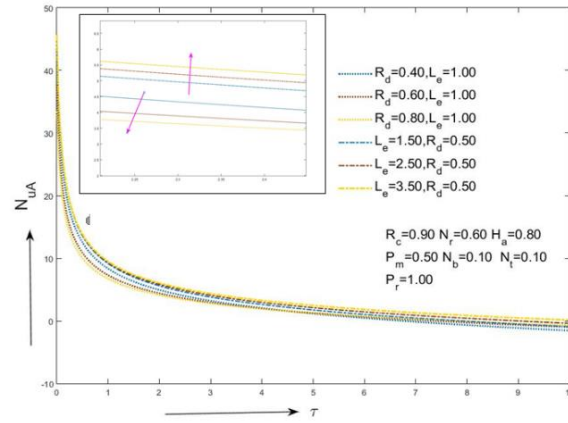


Figure 11. Effect of R_d , and L_e on average Nusselt number

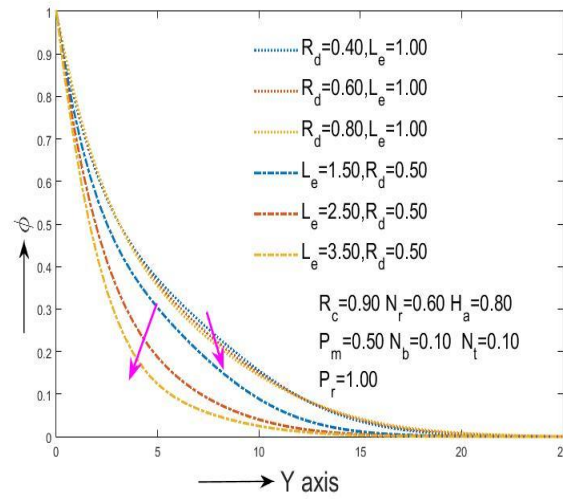


Figure 12. Effect of R_d , and L_e on the concentration distribution

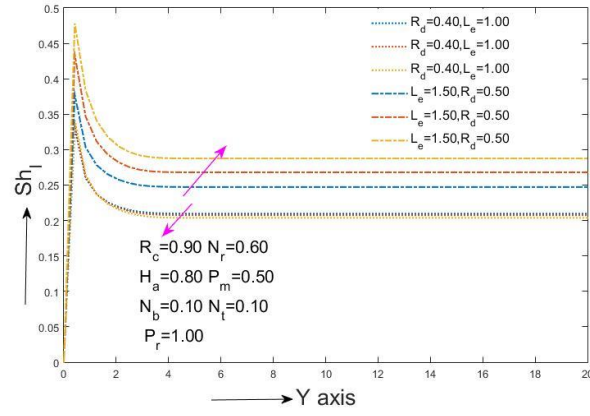


Figure 13. Effect of R_d , and L_e on local Sherwood number

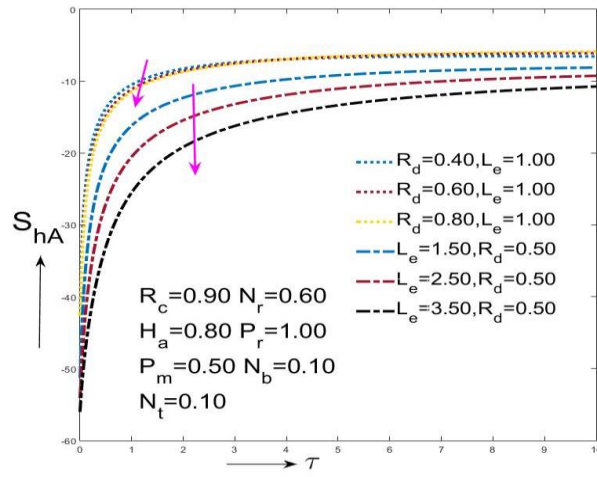


Figure 14. Effect of R_d , and L_e on average Sherwood number

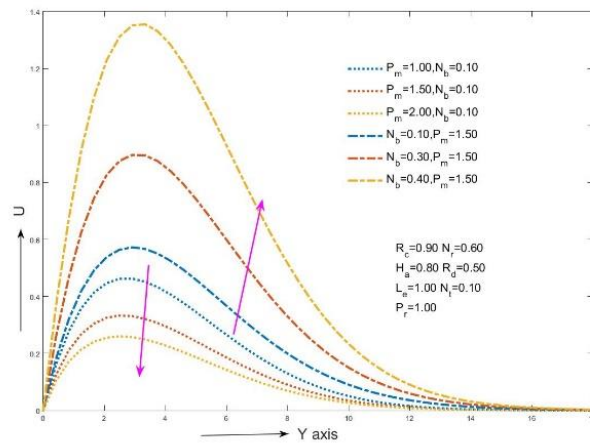


Figure 15. Effect of P_m , and N_b on velocity distribution

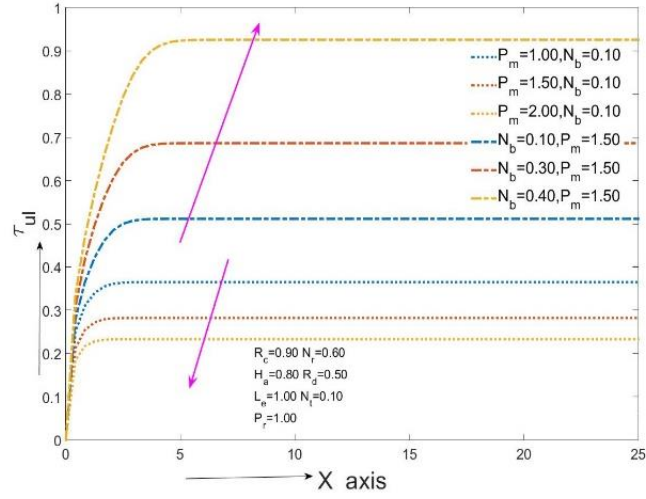


Figure 16. Effect of P_m , and N_b on local shear stress

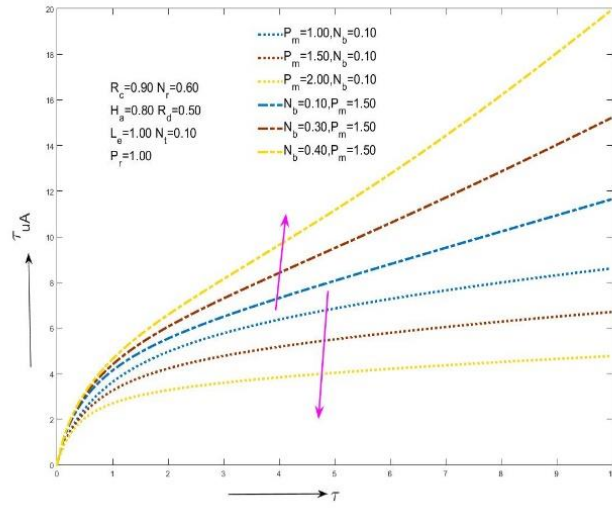


Figure 17. Effect of P_m , and N_b on average shear stress

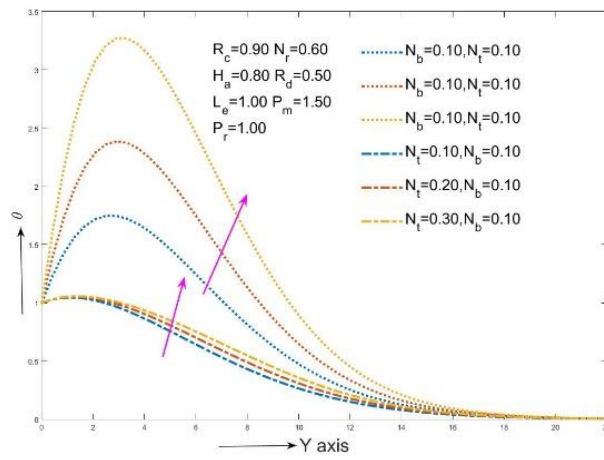


Figure 18. Effect of N_b , and N_t on temperature distribution

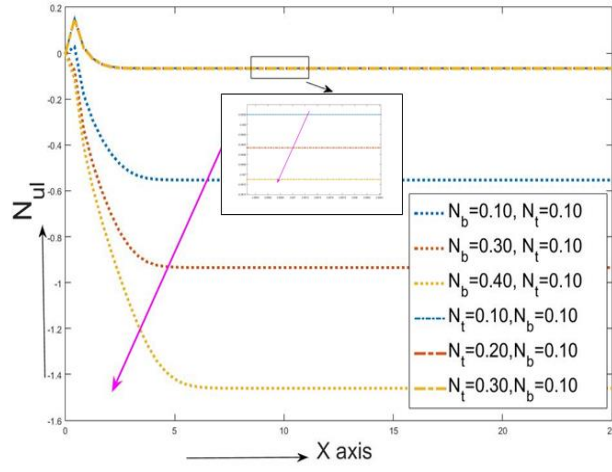


Figure 19. Effect of N_b , and N_t on local Nusselt number

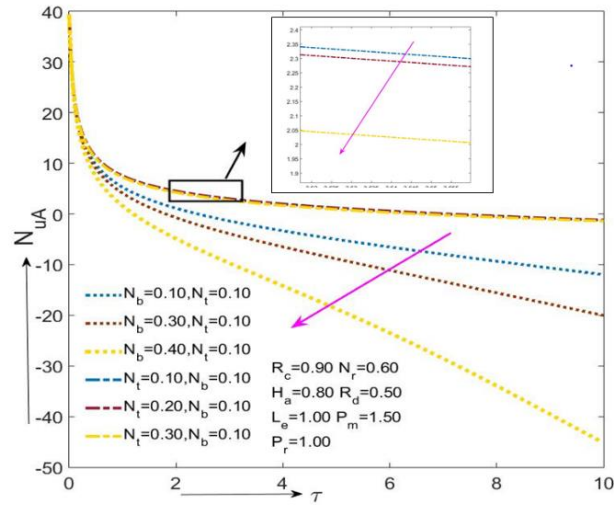


Figure 20. Effect of N_b , and N_t on average Nusselt number

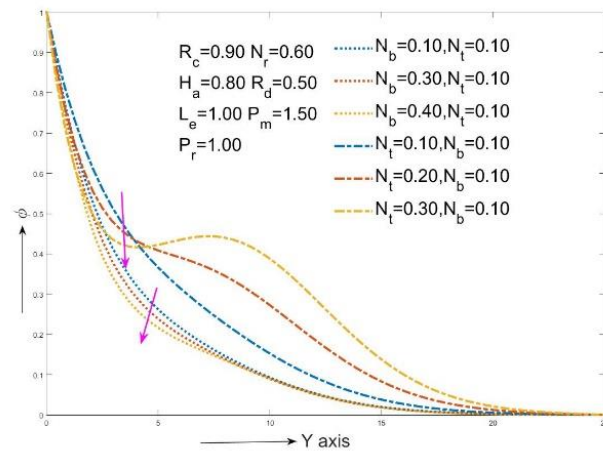


Figure 21. Effect of N_b , and N_t on concentration distribution

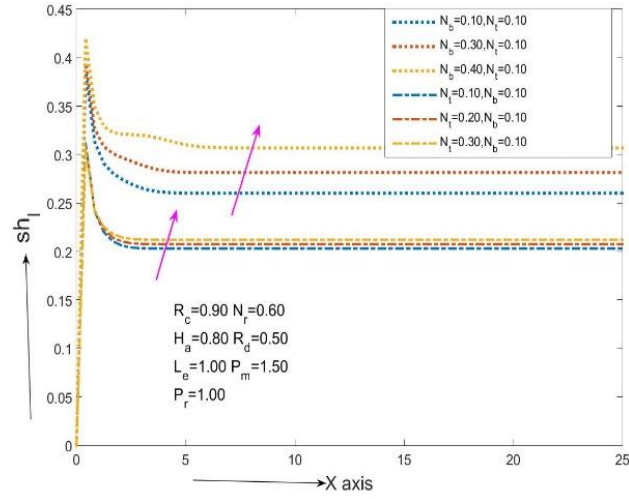


Figure 22. Effect of N_b , and N_t on local Sherwood number

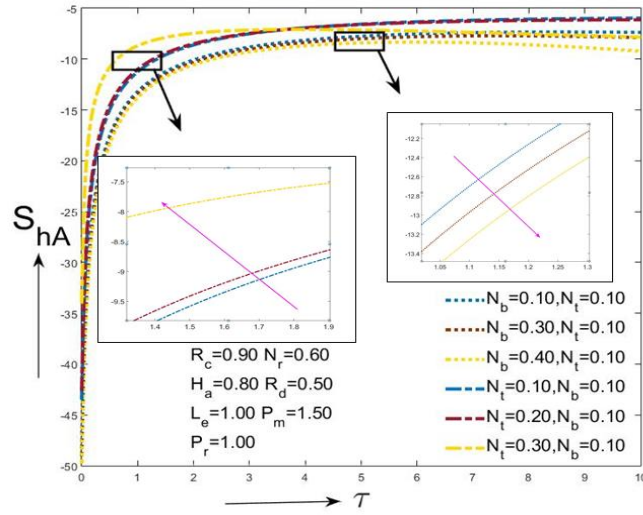


Figure 23. Effect of N_b , and N_t on average Sherwood number

CONCLUSION

An unsteady two-dimensional flow model is represented to investigate the explicit solutions with FDM for radiating nanofluid flowing through porous Riga plate. The results for the present study are expounding for several emerging parameters such as radiation parameter R_d , Lewis number L_e , porous medium parameter P_m , Brownian motion parameter for N_b and thermophoresis parameter N_t . The salient marks of the reported results are concluded below:

The velocity, and average shear stress hike up for the rising values of R_d and L_e while local shear stress decreases with R_d .

Temperature distribution and average Nusselt number decrease for L_e and increase for R_d , but local Nusselt number increases for both R_d and L_e .

Average Sherwood number along with the concentration fall for the increment of R_d and L_e whilst the local Sherwood number increases for the rise of L_e and decreases for R_d .

Velocity distribution, local shear stress, and average shear stress rise with the proportional rate of N_b and P_m while decrease with the increment of N_b and P_m .

Temperature distribution increases for both the rising values of N_b and N_t on the other hand local and average Nusselt numbers decline with the increment of N_b and N_t . Concentration and local Sherwood number increase with the rising values of N_b and N_t while average Sherwood number decreases for N_b but increases with the increment of N_t .

REFERENCES

- [1] S.U.S. Choi, "Enhancing thermal conductivity of fluids with nanoparticles", In the Proceedings of the ASME Int. Mech.Eng.Cong. and Exp., San Francisco, USA, ASME, FED 231/MD, Vol. 66, Pp. 99-105, 1995.
- [2] A. Gailitis, O. Lielausis, "On a possibility to reduce the hydrodynamic resistance of Plate in an electrolyte", Journal of Applied Magnetohydrodyn, Vol. 12, Pp. 143–146, 1961.
- [3] M. Afridi, M. Qasim, and A. Hussanan., "Second Law Analysis of Dissipative Flow over a Riga Plate with Non-Linear Rosseland Thermal Radiation and Variable Transport Properties", Entropy, Vol. 20, No. 8, Pp. 615, 2018. doi:10.3390/e20080615
- [4] E. Grinberg, "On the determination of properties of some potential fields", Journal of Applied Magnetohydrodynamics, Rep. Phys. Inst. Riga., Vol. 12, Pp. 147–154, 1961.
- [5] T. Abbas, M.M. Bhatti, and M. Ayub, "Aiding and opposing of mixed convection Casson nanofluid flow with chemical reactions through a porous Riga plate", Proceedings of the Institution of Mechanical Engineers, Part E: Journal of Process Mechanical Engineering, Vol. 232, No. 5, Pp. 519-527, 2017. doi:10.1177/0954408917719791
- [6] T. Abbas, M. Ayub, M.M. Bhatti, M.M. Rashidi, and M.E.S. Ali, "Entropy Generation on Nanofluid flow through a Horizontal Riga plate", Journal of MDPI, Vol. 18, No. 6, Pp. 223, 2016. <https://doi.org/10.3390/e18060223>
- [7] Y. Wang, T. Hayat, "Fluctuating flow of a Maxwell fluid past a porous plate with variable suction", Nonlinear Analysis. Real World Application, Vol. 9, No. 4, Pp. 1269-1282, 2008. <https://doi.org/10.1016/j.nonrwa.2007.02.016>
- [8] A. Ahmad, S. Asghar, and S. Afzal, "Flow of nanofluid past a Riga plate", Journal of Magnetism and Magnetic Materials, Vol. 402, Pp. 44-48, 2016. <https://doi.org/10.1016/j.jmmm.2015.11.043>
- [9] T. Hayat, M. Khan, M. Imtiaz, and A. Alsaedi, "Squeezing flow past a Riga plate with chemical reaction and convective conditions", Journal of Molecular Liquids, Vol. 225, Pp. 569-576, 2017. <https://doi.org/10.1016/j.molliq.2016.11.089>
- [10] Z. Iqbal, Z. Mehmood, E. Azhar, and E.N. Maraj, "Numerical investigation of nanofluidic transport of gyrotactic microorganisms submerged in water towards Riga plate", Journal of Molecular Liquids, Vol. 234, Pp. 296-308, 2017. <https://doi.org/10.1016/j.molliq.2017.03.074>
- [11] M.T. Mollah, S. Poddar, M.M. Islam, and M.M. Alam, 'Non-isothermal Bingham fluid flow between two horizontal parallel plates with Ion-slip and Hall currents', SN Applied Sciences, Vol. 3, No. 115, 2021. <https://doi.org/10.1007/s42452-020-04012-2>
- [12] S.U. Khan, K. Al-Khalid, and M.I. Khan, "Convective nonlinear thermally developed flow of thixotropic nanoliquid configured by Riga surface with gyrotactic microorganism and activation energy: A bio-technology and thermal extrusion model", International Communication in Heat and Mass Transfer, Vol. 111, Pp. 104966, 2020. <https://doi.org/10.1016/j.icheatmasstransfer.2020.104966>

- [13] S. Khatun, M.M. Islam, M.T. Mollah, S. Poddar, M.M. Alam "EMHD radiating fluid flow along a vertical Riga plate with suction in a rotating system", SN Applied Sciences, Vol. 3, No. 452, 2021. <https://doi.org/10.1007/s42452-021-04444-4>
- [14] M. Ramzan, M. Bilal, and J.D. Chung, "Radiative Williamson nanofluid flow over a convectively heated Riga plate with chemical reaction-A numerical approach", Chinese Journal of Physics, Vol. 55, No. 4, Pp. 1663-1673, 2017. <https://doi.org/10.1016/j.cjph.2017.04.014>
- [15] G. Rasool, and T. Zhang, "Characteristic of chemical reaction and convective boundary conditions in Powell-Eyring nanofluid flow along a radiative Riga plate", Heliyon, Vol. 5, No. 4, 2019. <https://doi.org/10.1016/j.heliyon.2019.e01479>
- [16] M. Shamshuddin, S.R. Mishra, O.A. Bég, and A. Kadir, "Viscous dissipation and Joule heating effects in non-Fourier MHD squeezing flow, heat and mass transfer between riga plates with thermal radiation: variational parameter method solutions", Arabian Journal for Science and Engineering, Vol. 44, Pp. 8053–8066, 2019. <https://doi.org/10.1007/s13369-019-04019-x>
- [17] A. Nassem, A. Shafiq, L. Zhao, and M.U. Farooq, "Analytical investigation of third grade nanofluidic flow over a riga plate using Cattaneo-Chirstov model", Results in Physics, Vol. 9, Pp. 961-969, 2018. <https://doi.org/10.1016/j.rinp.2018.01.013>
- [18] G.K. Ramesh, and B.J. Gireesha, "Nonlinear radiative flow of nanofluid past a moving/stationary Riga plate", Frontiers in Heat and Mass Transfer, Vol. 9, No. 1, Pp. 1-7, 2017. <http://dx.doi.org/10.5098/hmt.9.3>
- [19] A. Shafiq, I. Zari, I. Khan, T.S.Khan, A.H. Sheikh, E.S.M. Sherif, "Marangoni driven boundary layer flow of carbon nanotubes toward a Riga plate", Frontiers in Physics, Vol. 7, No. 215, 2020. <https://doi.org/10.3389/fphy.2019.00215>
- [20] A. Anjum, N.A. Mir, M. Farooq, M.I. Khan, and T. Hayat, "Influence of thermal stratification and slip conditions on stagnation point flow towards variable thicked Riga plate", Results in Physics, Vol. 9, Pp. 1021-1030, 2018. <https://doi.org/10.1016/j.rinp.2018.02.069>
- [21] T. Abbas, T., Bhatti, and M. Ayub, "Electromagnetohydrodynamic nanofluid flow past a porous riga plate containing gyrotactic microorganism", Neural Computing and Applications, Vol. 31, Pp. 1905–1913, 2017. <https://doi.org/10.1007/s00521-017-3165-7>
- [22] K. Ramesh, and U.S. Khan, "Bioconvection assessment in Maxwell nanofluid configured by a Riga surface with nonlinear thermal radiation and activation energy", Journal of Surfaces and Interfaces of Materials, Vol. 21, No. 18, Pp. 100749, 2020.
- [23] T. Thumma, M.V. Magagula, "Transient electro magneto hydrodynamic radiative squeezing flow between two parallel Riga plates using a spectral local linearization approach", Heat Transfer -Asian Research, Vol. 49, No. 1, Pp. 67-85, 2019. <http://dx.doi.org/10.1002/htj.21599>
- [24] P. Loganathan, and K. Deepa, "Boundary layer control of a radiative casson fluid flow past a permeable Riga plate with unimolecular chemical reaction", Latin American applied Research Pesquisa aplicada Latinoamericana, Vol. 49, No. 04, Pp. 233-239, 2019. <http://dx.doi.org/10.52292/j.laar.2019.135>
- [25] M.T. Mollah, "EMHD Laminar Flow of Bingham Fluid Between tow parallal Riga plates", International Journal of Heat and Technology, Vol. 37, No. 02, Pp. 641-648, 2019. <https://doi.org/10.18280/ijht.370236>
- [26] Y. Chu, F. Ahmad, and M.L. Khan, "Numerical and scale analysis of non-Newtonian fluid (Eyring-Powell) through pseudo-spectral collocation method (PSCM) towards a magnetized stretchable Riga surface", Alexandria Engineering Journal, Vol. 60, No. 2, Pp. 2127-2137, 2021. <http://dx.doi.org/10.1016/j.aej.2020.12.017>
- [27] S.M. Ansari, M.V. Magagula, and M. Trivedi, "Jeffrey nanofluid flow near a Riga plate: Spectral quasilinearization approach", Heat Transfer -Asian Research, Vol. 49, No. 3, Pp. 1491-1510, 2020. <https://doi.org/10.1002/htj.21673>

- [28] S. Poddar, M.M. Islam, J. Ferdouse, and M.M. Alam, “Characteristical analysis of MHD heat and mass transfer dissipative and radiating fluid flow with magnetic field induction and suction”, SN Applied Sciences, 3(470), 2021. <https://doi.org/10.1007/s42452-021-04452-4>
- [29] D.S. Chauhan, and V. Khemchandani, “Entrophy generation in the poiseuille flow of a temperature dependent viscosity fluid through a channel with a naturally permeable wall under thermal radiation”, Advances in Applied Science Research, Vol. 7, No. 4, Pp. 104-120, 2016.
- [30] A.R. Hassan, R. Maritz, and J.A. Gbadeyan, “A reactive hydromagnetic heat generating fluid flow with thermal radiation within porous channel with symmetrical convective cooling”, International Journal of Thermal Sciences, Vol. 122, Pp. 248-256, 2017. <https://doi.org/10.1016/j.ijthermalsci.2017.08.022>